PORTGEN

Dynamic Online Portfolio Generator

Project type: School software system documentation

Live URL: Not deployed

Target audience: Students, freelancers, and job seekers

Author: PortGen project implementation

Date: 2026-10-19

Version: 0.1.0

PortGen is a Laravel application for building and publishing a personal portfolio from one structured form. This paper documents the implemented system, the evidence gathered from automated checks and expert inspection, and the current deployment limit. The code, local screenshots, and design decisions are available in this repository. No classmates have taken part in the usability study yet, and there is no live deployment.

1. System overview

1.1 Problem and solution

Students and early-career professionals often have information spread across documents and profile sites. A portfolio can be time-consuming to format and keep consistent. PortGen collects a person's introduction, education, work experience, skills, projects, and social links in one guided form. The person chooses a responsive presentation and receives a public portfolio URL plus a separate signed link for private management.

1.2 Objectives

1. Collect structured portfolio information in a guided, recoverable flow.
2. Render the same information with Simple, Modern, or Creative presentation styles.
3. Let a portfolio owner edit, change the template, and delete the portfolio without requiring an account.
4. Support keyboard operation, responsive layouts, clear feedback, and a dark color scheme.
5. Keep database constraints, request validation, automated browser checks, and HCI evidence in the project repository.

1.3 Scope and limits

The implementation includes portfolio creation, template selection, public viewing, signed-link management, editing, deletion with a ten-second undo window, friendly error pages, a demo portfolio, CI configuration, and Docker/Render deployment configuration. It does not include accounts, password recovery, image uploads, payments, analytics, or user research results. The public demo contains fictional information.

The deployment is not approved as production-ready. The required Laravel 11 line is past its published security-fix window, and `composer audit --locked` reports four advisories for the locked dependency set, including a medium-severity signed-URL path-confusion advisory. Docker and production database and Render configuration were unavailable in the workspace. The application was not deployed and the PostgreSQL CI workflow was not run from this checkout. See section 8 for the exact release conditions.

2. Users and needs

2.1 Personas and tasks

Table 1. Target users and primary needs

Persona	Situation and tasks	Likely needs
Student	Building an early-career profile with limited work history; adds education, skills, and coursework projects, selects a template, and shares the page.	Examples for unfamiliar fields, useful empty states, and a way to show projects without paid experience.
Freelancer	Presents services and selected client work; edits the portfolio as work changes and shares it with prospects.	Fast editing, a work-first layout, and a private owner link that can be saved.
Job seeker	Prepares a focused profile for applications; selects a readable layout and revises details.	Straightforward form, keyboard access, clear completion feedback, and predictable editing and deletion.

These are design personas and assumptions, not research findings. The main assumptions are that users can provide the information in one sitting if examples are available, that the public and private links are understandable, and that three initial templates provide enough variety. The study kit in `docs/usability-testing/` leaves participant outcomes open for classmates.

2.2 User-needs walkthrough

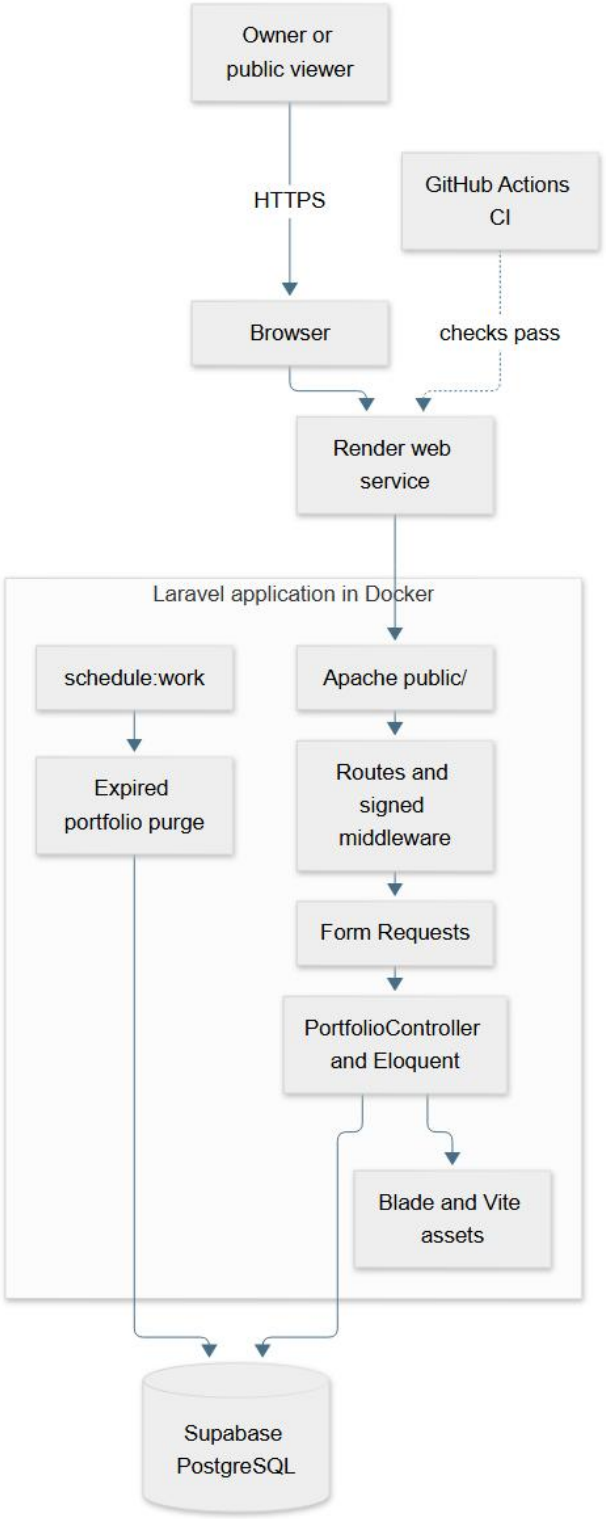
The student path permits empty education and work sections while requiring at least one skill. This supports users with limited employment history while ensuring that a portfolio has at least one concrete strength. The freelancer path supports repeatable project entries and an owner link for later changes. The job-seeker path uses familiar labels, inline validation, a completion stage, and a confirmation before deletion. These are expert walkthrough conclusions to validate with participants.

3. System architecture and technology stack

3.1 Architecture

The browser submits validated portfolio fields to Laravel. The controller and Eloquent model persist structured values in PostgreSQL JSONB columns. Blade renders the public page from the chosen template. A signed URL provides private owner access without an account. A Laravel scheduler in the container permanently removes records after their undo deadline.

Figure 1. Application architecture



The user-facing application is a modular Laravel monolith. The controller handles the portfolio workflow, Form Requests validate input, signed middleware protects owner routes, Blade templates render screens, and the database stores portfolio records. The runtime scheduler is supervised alongside Apache by `docker/start.sh`.

3.2 Technology stack

Table 2. Implemented and planned technology

Layer	Technology	Use and status
Server framework	Laravel 11, PHP 8.3 target	MVC routing, validation, ORM, migrations, signed URLs, and scheduled commands. The local workstation runs PHP 8.5.
Views and styling	Blade, Tailwind CSS 4, Vite 8	Server-rendered views, design tokens, responsive styling, and production asset bundles.
Database	PostgreSQL 16 in CI; Supabase PostgreSQL planned for production	PostgreSQL is configured as a CI service. No production database is connected. Local tests use isolated SQLite.
Browser checks	Playwright 1.64 and axe-core 4.14	Keyboard journey, responsive screenshots, touch target checks, and accessibility audits.
Container	Docker, PHP 8.3 Apache	Multi-stage image configuration is committed; image build was not run because Docker is unavailable here.
Hosting	Render Blueprint	Web service and health-check configuration are committed; no service was created.
CI	GitHub Actions	Workflow defines PHP, PostgreSQL, build, browser, and HCI artifact steps. It has not run on GitHub.

Table 3. Main workflow stages

Stage	User action	System response
1	Enter portfolio details	Form Request validates fields; the app stores a draft locally and shows inline feedback.
2	Compare and choose a template	The app previews the user's information and saves the selected supported template through a signed route.
3	Publish and share	The app displays the public URL and a private signed management URL with a copy action.
4	Edit or change a template	The owner opens the private signed link and updates the same portfolio record.
5	Delete or undo	The app soft-deletes the record, allows signed recovery for ten seconds, then purges expired rows on the server schedule.

4. Database schema and data dictionary

4.1 Portfolio entity

Figure 2. Portfolio data model



There is one application-owned `portfolios` entity. The structured content fields store arrays as JSONB in PostgreSQL and are cast to arrays by Eloquent. The table has no foreign-key relationship because portfolios do not require user accounts.

Table 4. `portfolios` data dictionary

Column	PostgreSQL type	Constraint	Description
id	uuid	Primary key	Application-generated portfolio identifier.
slug	varchar(10)	Unique, required	Random ten-character route key used in public URLs.
full_name	varchar(120)	Required	Name displayed on the portfolio.
headline	varchar(180)	Required	Short professional tagline.

Column	PostgreSQL type	Constraint	Description
about_me	text	Required	Introduction, limited to 2,000 characters by request validation.
education	jsonb	Required array	Up to ten education entries with institution, qualification, and year.
work_experience	jsonb	Required array	Up to ten work entries with company, role, description, and month fields.
skills	jsonb	Required array, 1-30 entries	Skill names, up to 50 characters each.
projects	jsonb	Required array	Up to twenty entries with name, description, and optional HTTP/HTTPS URL.
social_links	jsonb	Required array	Up to twenty labeled HTTP/HTTPS links.
template_choice	varchar	Nullable; check constraint	Null before selection, otherwise Simple, Modern, or Creative.
created_at	timestamp	Nullable	Laravel creation timestamp.
updated_at	timestamp	Nullable	Laravel update timestamp.
deleted_at	timestamp	Nullable	Soft-delete marker; the record remains recoverable during the undo period.
undo_until	timestamp	Nullable	Deadline after which the scheduled purge may permanently delete the row.

PostgreSQL enforces the template choice with a CHECK constraint. SQLite migration tests use insert/update triggers to enforce the same allowed values. `Portfolio` has an explicit `$fillable` list; IDs, slugs, timestamps, and deletion fields cannot be mass assigned from request input. The factory creates test data and the idempotent seeder creates the fictional `portgen001` demo.

5. Key routes and controller logic

5.1 Route map

Table 5. Main HTTP routes

Method	URL	Purpose	Access
GET	/	Redirect to the portfolio form	Public
GET	/portfolios/create	Open the creation flow	Public
POST	/store	Validate and create the portfolio	Public, rate limited
GET	/portfolio/{slug}/select-template	Choose a presentation	Signed
POST	/portfolio/{slug}/save-template	Save an allowed template	Signed, rate limited
GET	/p/{slug}	Display the public portfolio	Public

Method	URL	Purpose	Access
GET	/manage/{slug}	Open owner controls	Signed
GET	/manage/{slug}/edit	Edit the portfolio	Signed
PUT	/manage/{slug}	Validate and update content	Signed, rate limited
DELETE	/manage/{slug}	Begin the delete and undo flow	Signed, rate limited
GET	/manage/{slug}/deleted	Display the deletion countdown	Signed
POST	/manage/{slug}/undo	Restore during the undo window	Signed, rate limited
DELETE	/manage/{slug}/purge	Purge after the undo period	Signed, rate limited
GET	/help	Read help and keyboard shortcuts	Public
GET	/up	Health check	Public health endpoint

Error preview routes exist only in local and test environments. The production routes return friendly 403, 404, 419, 429, and 500 views. State-changing routes retain CSRF protection and use Laravel throttling.

5.2 Controller and validation

`PortfolioController` implements create, store, template selection, template save, public show, manage, edit, update, delete, deleted-state display, undo, and purge. `StorePortfolioRequest` and `UpdatePortfolioRequest` share portfolio rules. `SaveTemplateRequest` restricts template values. URLs accept only HTTP and HTTPS schemes. User content is rendered through escaped Blade expressions.

Owner management is a bearer-link model. Laravel signed URLs without an expiry protect owner routes; anyone who obtains the private link can manage that portfolio. The completion screen explains the link, shows the public URL separately, and provides a copy control. There is no account-based recovery if the owner loses the private link.

Security middleware adds Content Security Policy, `X-Frame-Options`, `X-Content-Type-Options`, `Referrer-Policy`, and `Permissions-Policy`. `APP_DEBUG` is set to false in the Render configuration. Production also requires a real `APP_KEY` in the hosting secret store.

6. User interface design and HCI compliance

6.1 Interaction design

The creation flow uses three named stages and collapsible form sections. The browser stores recoverable draft fields in localStorage, excluding CSRF and method fields. A restore prompt appears on reload. Inline errors identify missing values; submit, copy, and deletion provide visible status. Users can operate the core flow with pointer controls or keyboard navigation and can open the shortcut dialog from the shared layout.

The three public templates group the same information differently. Simple uses a quiet neutral palette, Modern uses neighboring blue hues, and Creative uses warm cream with plum and magenta accents. All colors are defined as CSS custom properties in one token block with dark-mode overrides. The test computes contrast for configured text and interface pairs in both modes. It is a token-level regression test; it is not a manual audit of every rendered color combination.

6.2 Accessibility evidence

The Playwright suite tests the full keyboard create-and-publish journey, skip-link behavior, inline validation, local draft restoration, template changes, copy feedback, signed management, edit, delete and undo, help, and error screens. For each of 17 captured states it runs axe-core at 375 by 812 and 1280 by 800, checks visible control target dimensions, and checks the mobile layout at 320 pixels for horizontal overflow. The create screen is additionally inspected at 200%

browser zoom. The automated suite reports no serious or critical axe violations. The suite does not claim complete WCAG conformance or replace assistive-technology user testing.

6.3 Screenshot evidence

Each of the 17 listed screen states has a mobile and desktop full-page image under `docs/screenshots/`. Captures use fictional profile content. The private signed owner URL is replaced in the saved screenshot with an invalid example value; the browser test still verifies that the actual link copies correctly before redaction. Each exported figure shows the 375 by 812 mobile viewport and 1280 by 800 desktop viewport crop, with the full-page captures retained in the repository. Each pair is annotated with its principal HCI evidence.

6.3.1 Creation and template selection

The first three figures show the form, template previews, and completion screen. They provide evidence for staged task structure, field guidance, visible status, and preview-based recognition.

Figure 3. Create form, step 1

Mobile: `docs/screenshots/mobile/step-1-your-info.png`

Desktop: `docs/screenshots/desktop/step-1-your-info.png`

HCI evidence: S1, S5, S8, N1, N2, N3, N4, H2, H5, H8, U1, U3, U6, U7, V1, V2, V3, V7, V8.

Figure 4. Template selection, step 2

Mobile: `docs/screenshots/mobile/step-2-choose-template.png`

Desktop: `docs/screenshots/desktop/step-2-choose-template.png`

HCI evidence: N2, N5, N3, H6, U3, C1, V3.

Figure 5. Completion and sharing, step 3

Mobile: `docs/screenshots/mobile/step-3-done-share.png`

Desktop: `docs/screenshots/desktop/step-3-done-share.png`

HCI evidence: S3, S4, H1, U4, C4, V7. The owner link is redacted in the saved image.

6.3.2 Public portfolio templates

Figure 6. Public Simple template

Mobile: `docs/screenshots/mobile/template-simple.png`

Desktop: `docs/screenshots/desktop/template-simple.png`

HCI evidence: S1, H4, H8, C1, C2, C3, V1, V2, V4, V5, V8.

Figure 7. Public Modern template

Mobile: `docs/screenshots/mobile/template-modern.png`

Desktop: `docs/screenshots/desktop/template-modern.png`

HCI evidence: H6, C1, C2, C3, V3, V5, V6.

Figure 8. Public Creative template

Mobile: `docs/screenshots/mobile/template-creative.png`

Desktop: `docs/screenshots/desktop/template-creative.png`

HCI evidence: H6, C1, C2, C3, V4, V6, V8.

Figure 9. Creative template in dark mode

Mobile: `docs/screenshots/mobile/dark-mode-creative.png`

Desktop: `docs/screenshots/desktop/dark-mode-creative.png`

HCI evidence: C2, C3, V5. The CSS contrast test verifies configured palette pairs in light and dark mode.

6.3.3 Owner management and recovery

Figure 10. Private manage screen

Mobile: `docs/screenshots/mobile/manage.png`

Desktop: `docs/screenshots/desktop/manage.png`

HCI evidence: S1, S7, H1, H3, H4, N7, V1, V8.

Figure 11. Edit screen

Mobile: `docs/screenshots/mobile/edit.png`

Desktop: `docs/screenshots/desktop/edit.png`

HCI evidence: S1, N4, H3, U2, V2, V8.

Figure 12. Delete and undo

Mobile: `docs/screenshots/mobile/delete-undo.png`

Desktop: `docs/screenshots/desktop/delete-undo.png`

HCI evidence: S6, H3, H5, N6, U5. The undo window is ten seconds by requirement.

6.3.4 Help and errors

Figure 13. Help and FAQ

Mobile: `docs/screenshots/mobile/help-faq.png`

Desktop: `docs/screenshots/desktop/help-faq.png`

HCI evidence: N1, H2, H10, U1.

Figure 14. Keyboard shortcuts dialog

Mobile: `docs/screenshots/mobile/keyboard-shortcuts.png`

Desktop: `docs/screenshots/desktop/keyboard-shortcuts.png`

HCI evidence: S2, S3, H7, H10, U2, U6.

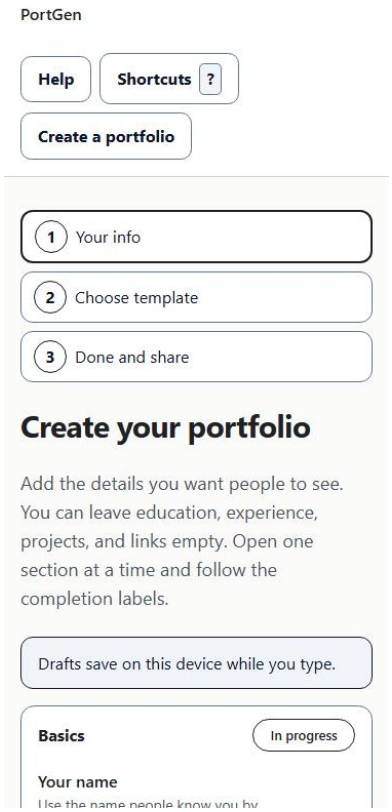
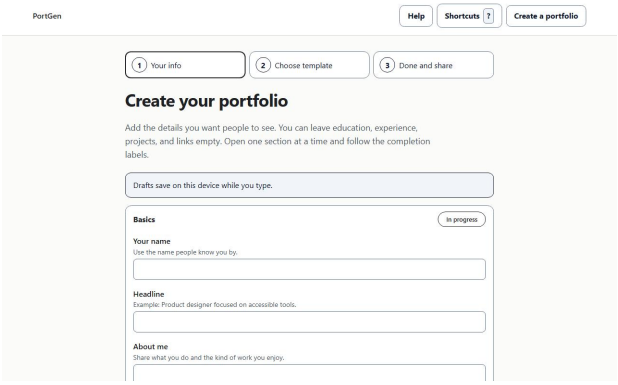
Figures 15-19. Friendly error states

For each error, the mobile and desktop captures appear in the exported paper: 403 `docs/screenshots/{mobile,desktop}/error-403.png`; 404 `error-404.png`; 419 `error-419.png`; 429 `error-429.png`; 500 `error-500.png`. The pages provide plain-language feedback and a return path. HCI evidence: S3, S5, N6, H1, H9, U4, C4.

6.3.5 Complete screenshot gallery

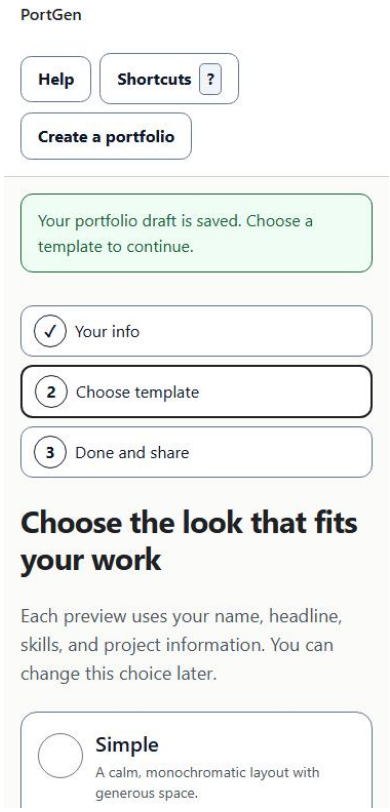
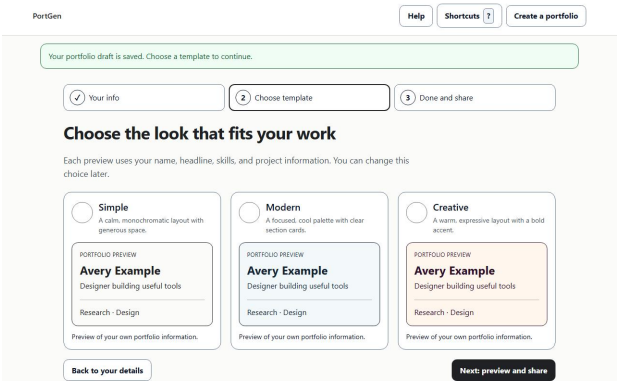
The following paired figures embed every mobile and desktop screen capture. Labels name the state shown and the primary principles illustrated.

Figure 20. Create form, step 1

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

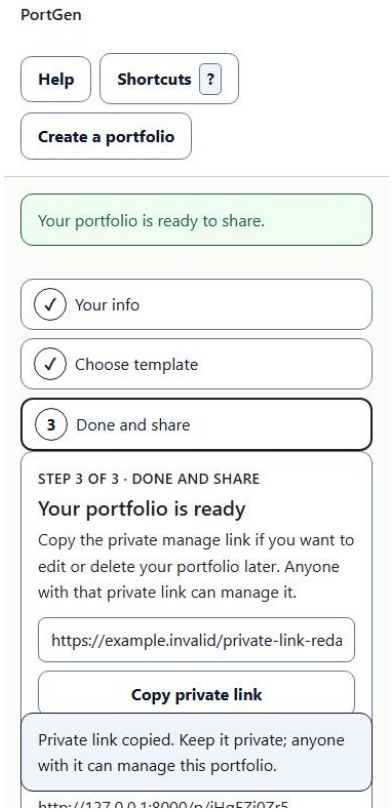
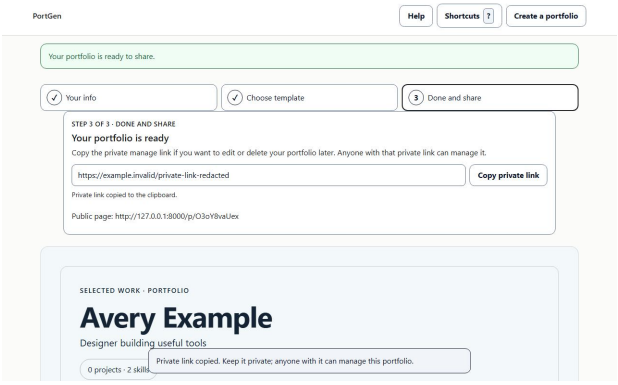
HCI principles: S8 Reduce short-term memory load; N4 Get the mappings right; H5 Error prevention; U7 Size and space for approach and use; V3 Visual hierarchy.

Figure 21. Template selection, step 2

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

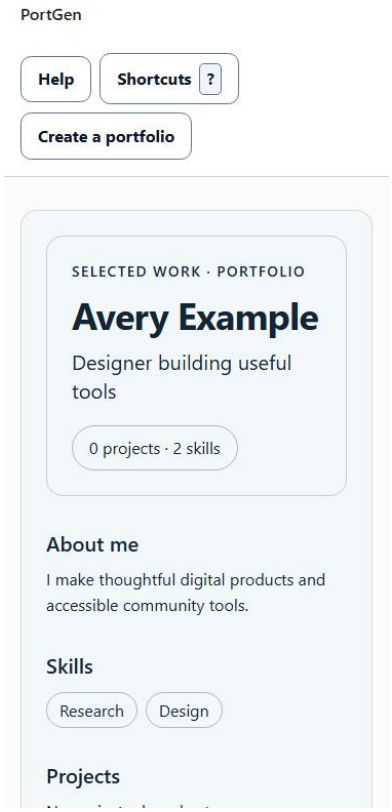
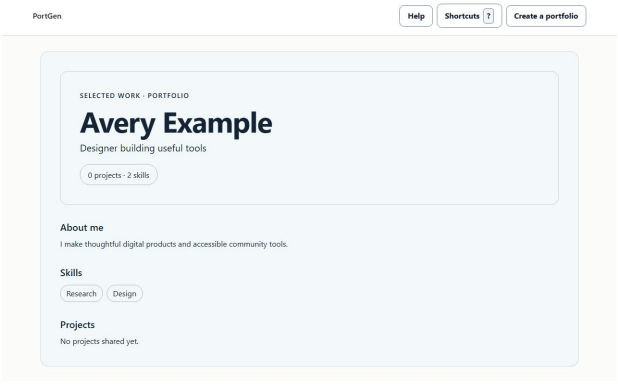
HCI principles: N2 Simplify task structure; H6 Recognition rather than recall; C1 Deliberate color scheme per template; V3 Visual hierarchy.

Figure 22. Completion and sharing, step 3

Mobile view	Desktop view
Mobile - 375 x 812  	Desktop - 1280 x 800 

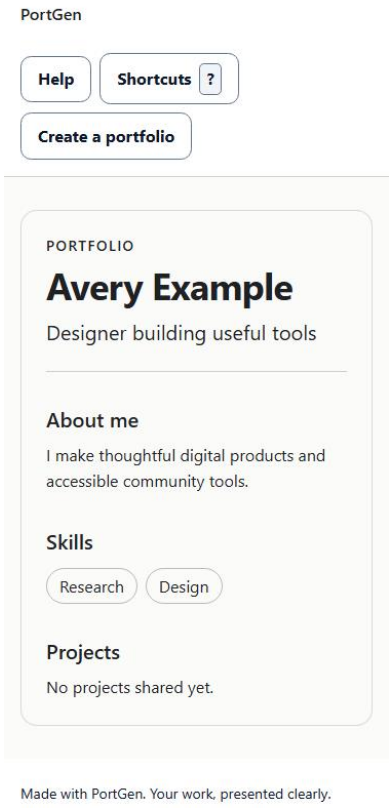
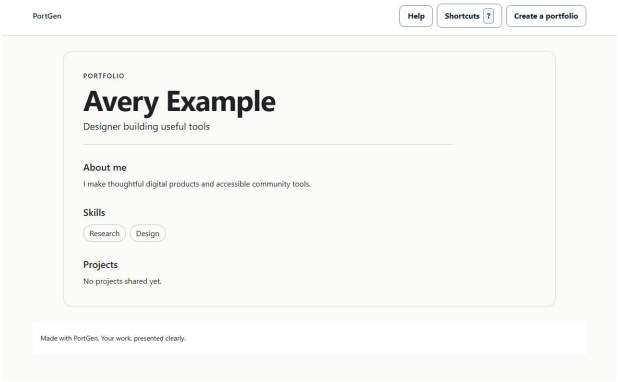
HCI principles: S3 Informative feedback; H1 Visibility of system status; C4 Never rely on color alone; V7 Dominance. the signed owner link is redacted in saved evidence

Figure 23. Modern portfolio preview

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

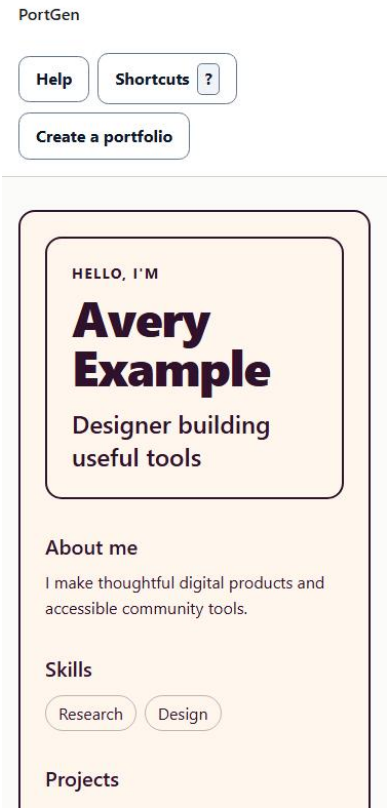
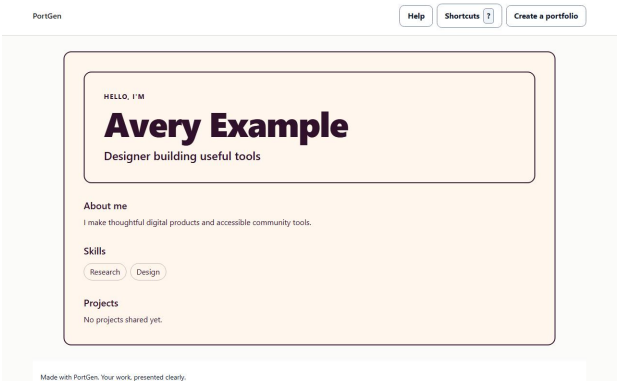
HCI principles: H6 Recognition rather than recall; C1 Deliberate color scheme per template; C3 Sufficient value contrast, WCAG AA; V5 Contrast.

Figure 24. Public Simple portfolio

Mobile view	Desktop view
Mobile - 375 x 812  	Desktop - 1280 x 800 

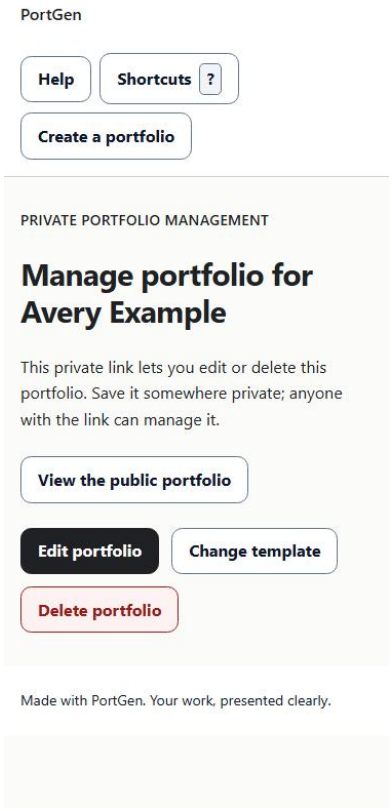
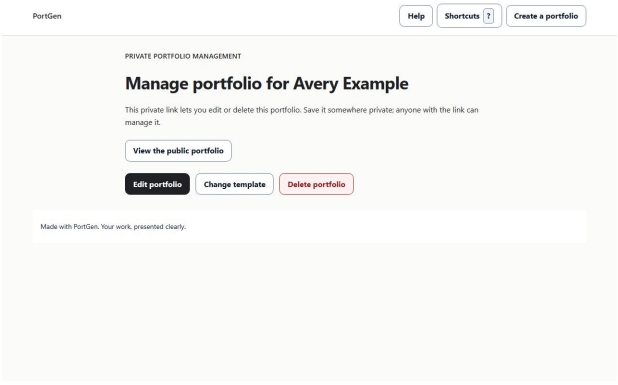
HCI principles: H8 Aesthetic and minimalist design; C1 Deliberate color scheme per template; V2 Gestalt principles; V4 Balance.

Figure 25. Public Creative portfolio

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

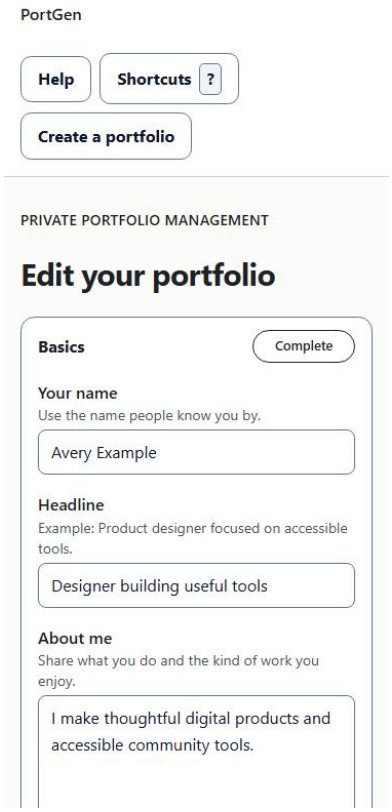
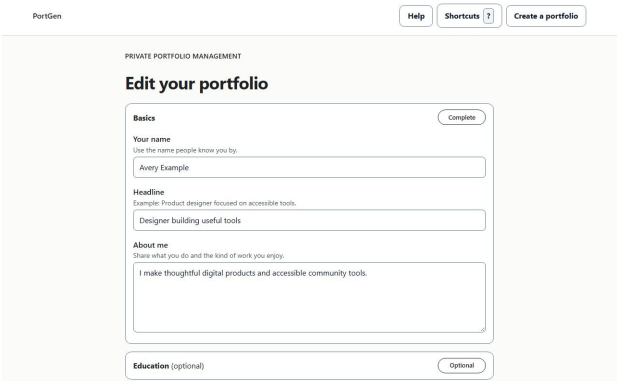
HCI principles: H6 Recognition rather than recall; C1 Deliberate color scheme per template; C2 Color temperature matches the template's message; V6 Scale.

Figure 26. Private portfolio management

Mobile view	Desktop view
Mobile - 375 x 812  	Desktop - 1280 x 800 

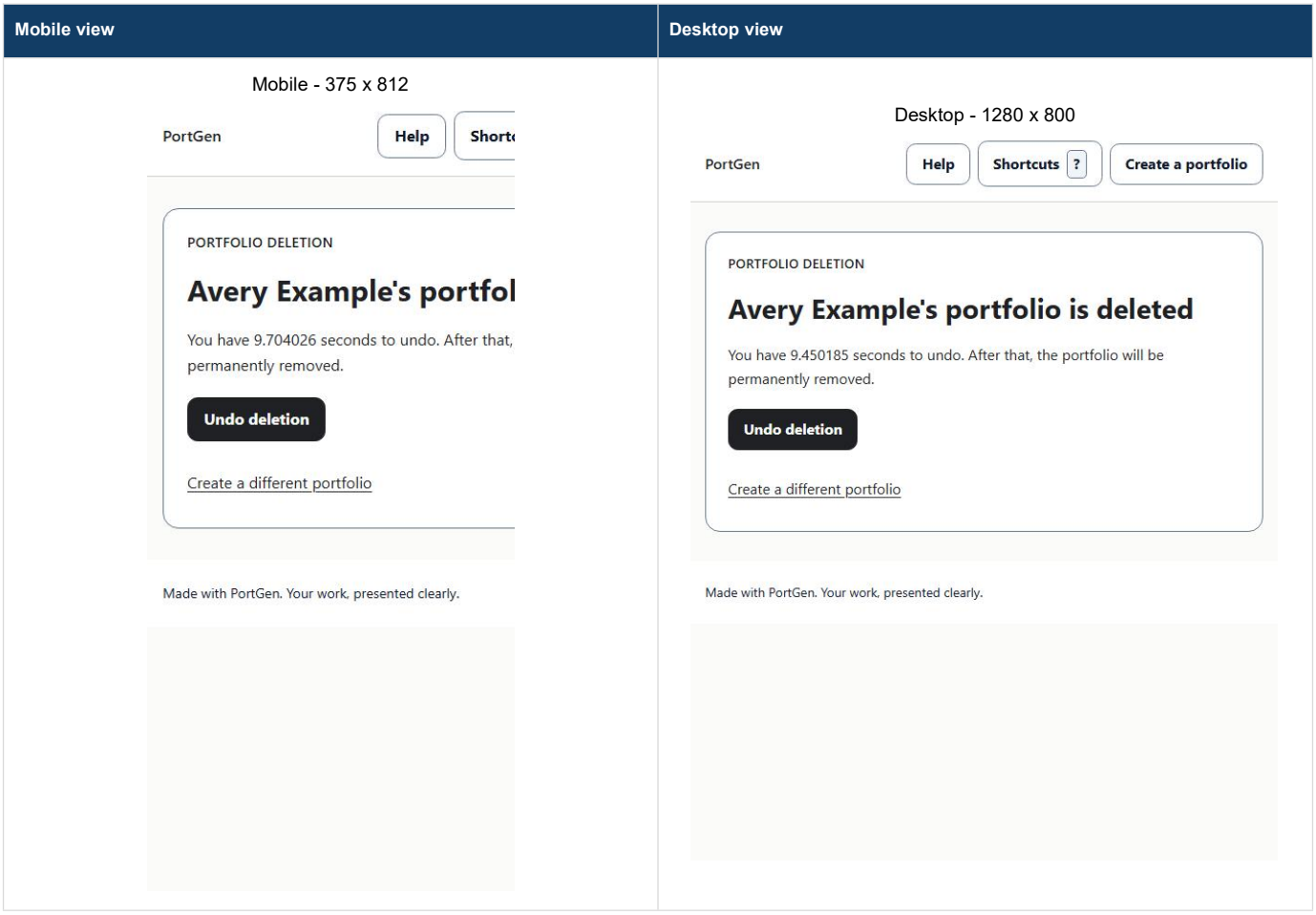
HCI principles: S7 Internal locus of control; H1 Visibility of system status; H3 User control and freedom; H4 Consistency and standards.

Figure 27. Edit portfolio

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

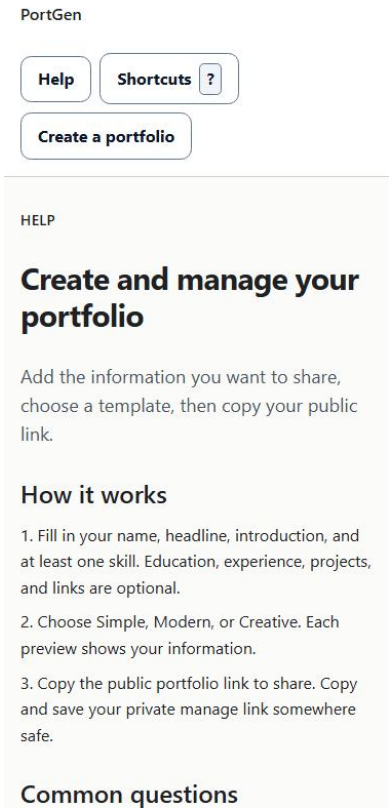
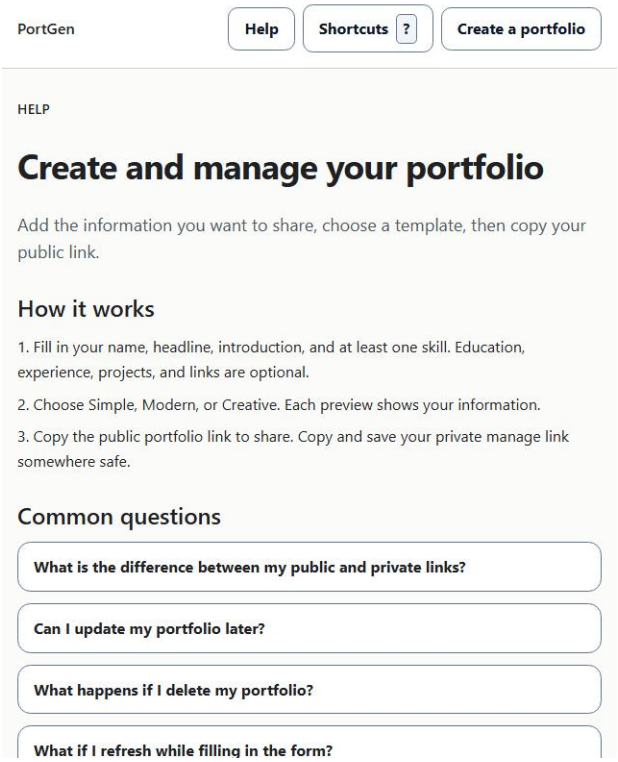
HCI principles: H3 User control and freedom; N4 Get the mappings right; U2 Flexibility in use; V2 Gestalt principles.

Figure 28. Delete and undo



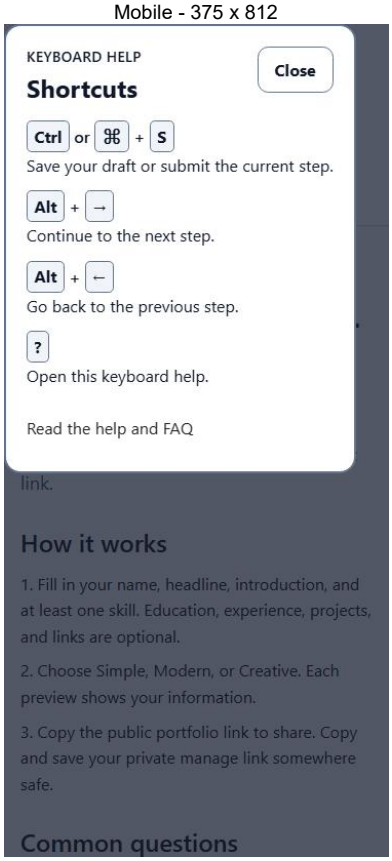
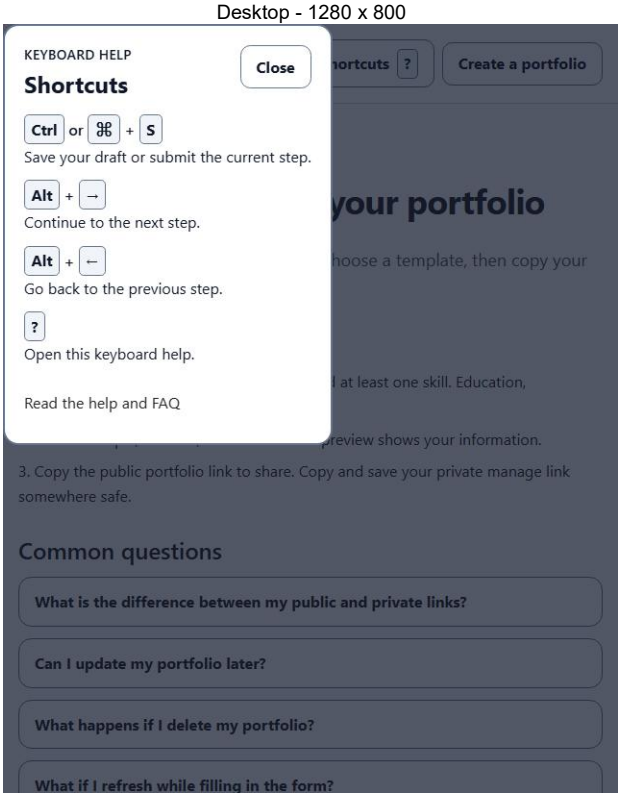
HCI principles: S6 Easy reversal of actions; H3 User control and freedom; U5 Tolerance for error. ten-second undo state

Figure 29. Help and FAQ

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: N1 Knowledge in the world and in the head; H10 Help and documentation; U1 Equitable use.

Figure 30. Keyboard shortcuts dialog

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: S2 Shortcuts for frequent users; H7 Flexibility and efficiency of use; U2 Flexibility in use.

Figure 31. Invalid private link, 403

Mobile view	Desktop view
Mobile - 375 x 812  	Desktop - 1280 x 800 

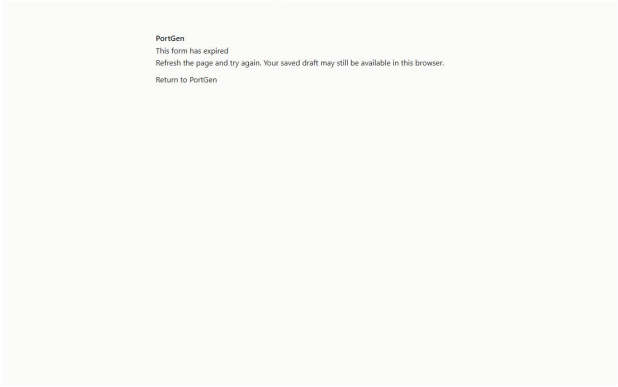
HCI principles: S3 Informative feedback; H9 Help users recognize, diagnose, recover from errors; C4 Never rely on color alone.

Figure 32. Missing portfolio, 404

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: S3 Informative feedback; H9 Help users recognize, diagnose, recover from errors; C4 Never rely on color alone.

Figure 33. Expired form, 419

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

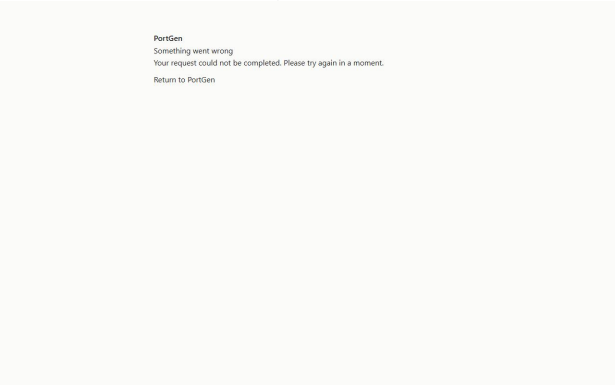
HCI principles: S3 Informative feedback; N6 Design for error; H9 Help users recognize, diagnose, recover from errors.

Figure 34. Rate limited, 429

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

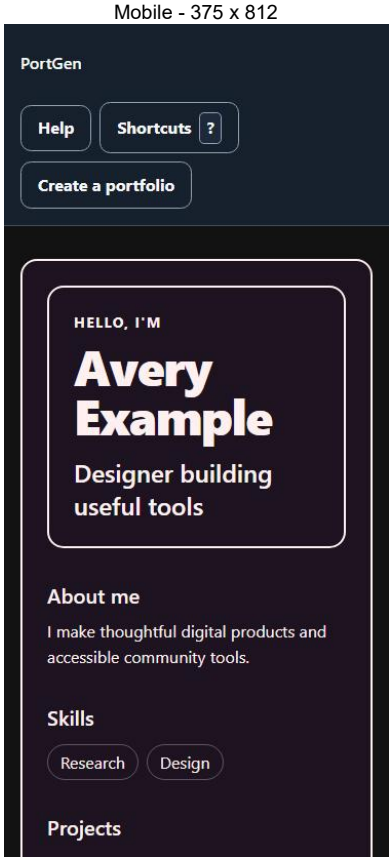
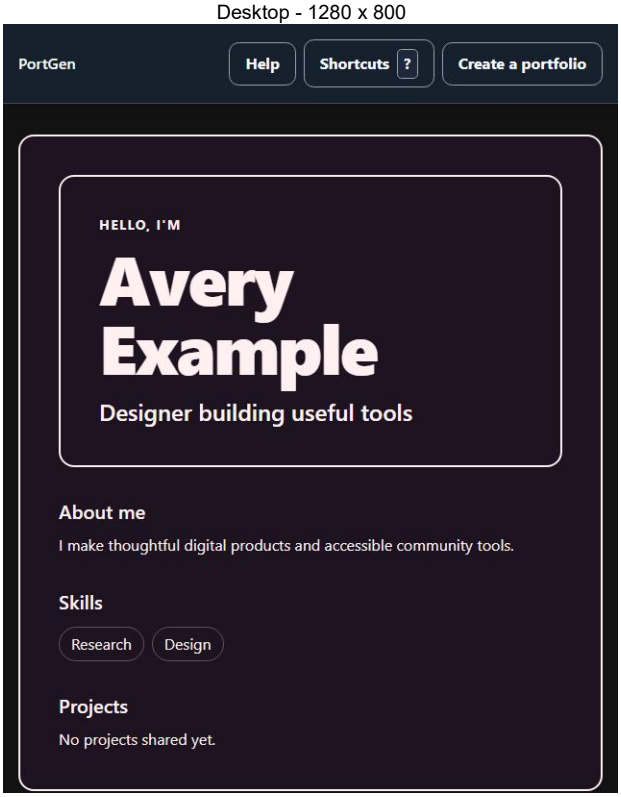
HCI principles: S3 Informative feedback; N6 Design for error; H9 Help users recognize, diagnose, recover from errors.

Figure 35. Server error, 500

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: S3 Informative feedback; N6 Design for error; H9 Help users recognize, diagnose, recover from errors.

Figure 36. Creative portfolio in dark mode

Mobile view	Desktop view
Mobile - 375 x 812 	Desktop - 1280 x 800 

HCI principles: C2 Color temperature matches the template's message; C3 Sufficient value contrast, WCAG AA; V5 Contrast.

6.4 Full HCI tracker

The tables below reproduce all 48 rows from `docs/HCI_TRACKER.md`. All 48 rows now have a named test or committed screenshot, and the `HCI_RELEASE=1` release test passes. `Verified` means the row cites implementation evidence; it does not mean every criterion has been tested with participants.

Shneiderman's 8 Golden Rules

Table 6. Shneiderman's 8 Golden Rules HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
S1	Consistency	resources/views/layouts/app.blade.php; resources/css/app.css	Shared layout, navigation, spacing, and repeated button styles.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/desktop/step-1-your-info.png	Verified
S2	Shortcuts for frequent users	resources/js/app.js; resources/views/layouts/app.blade.php	Discoverable save and step navigation shortcuts.	docs/screenshots/mobile/keyboard-shortcuts.png; docs/screenshots/desktop/keyboard-shortcuts.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified
S3	Informative feedback	resources/views/layouts/app.blade.php; resources/views/portfolios/show.blade.php	Announce saved, published, copied-link, and error status in polite live regions.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/desktop/step-3-done-share.png; PortfolioRoutesTest::test_template_selection_uses_and_preserves_the_private_link_for_the_creator	Verified
S4	Dialogs that yield closure	resources/views/portfolios/manage.blade.php; resources/views/portfolios/show.blade.php	Name the portfolio in the delete dialog and show the private owner link when creation finishes.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/desktop/step-3-done-share.png	Verified
S5	Error prevention and simple error handling	app/Http/Requests/Concerns/HasPortfolioRules.php; resources/views/portfolios/form.blade.php; resources/js/app.js	Validate structured fields and safe URLs, then show specific inline recovery messages.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; docs/screenshots/mobile/step-1-your-info.png	Verified
S6	Easy reversal of actions	routes/web.php; resources/views/portfolios/deleted.blade.php; app/Http/Controllers/PortfolioController.php	Offer a signed 10-second undo before permanent removal.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; docs/screenshots/mobile/delete-undo.png; docs/screenshots/desktop/delete-undo.png	Verified
S7	Internal locus of control	routes/web.php; app/Http/Controllers/PortfolioController.php	Require a private signed URL for management and keep transitions user initiated.	PortfolioRoutesTest::test_signed_manage_link_opens_and_unsigned_link_is_forbidden; docs/screenshots/mobile/manage.png	Verified
S8	Reduce short-term memory load	resources/views/portfolios/form.blade.php; resources/js/app.js	Group form fields, show completion labels, and offer local draft restoration.	docs/screenshots/mobile/step-1-your-info.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified

Norman's 7 Principles

Table 7. Norman's 7 Principles HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
N1	Knowledge in the world and in the head	resources/views/portfolios/form.blade.php; resources/views/help.blade.php	Use examples and plain-language help beside unfamiliar fields.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/step-1-your-info.png	Verified
N2	Simplify task structure	resources/views/portfolios/stepper.blade.php; resources/views/portfolios/create.blade.php; resources/views/portfolios/select-template.blade.php	Divide creation into a visible three-step flow.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/desktop/step-2-choose-template.png	Verified
N3	Make things visible	resources/views/portfolios/stepper.blade.php; resources/js/app.js; resources/views/layouts/app.blade.php	Show current step, section completion, save status, and submit loading state.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/mobile/step-3-done-share.png	Verified
N4	Get the mappings right	resources/views/portfolios/form.blade.php; resources/views/portfolios/templates/sections.blade.php	Use familiar field names and group entered information under matching headings.	docs/screenshots/mobile/step-1-your-info.png	Verified
N5	Use constraints	app/Http/Requests/SaveTemplateRequest.php; app/Http/Requests/Concerns/HasPortfolioRules.php	Restrict template choice and URL schemes before storage.	PortfolioRoutesTest::test_template_choice_is_limited_to_the_three_supported_values; docs/screenshots/mobile/step-2-choose-template.png	Verified
N6	Design for error	resources/js/app.js; resources/views/portfolios/deleted.blade.php	Autosave drafts, explain validation failures, and provide delete recovery.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window	Verified
N7	When all else fails, standardize	resources/views/layouts/app.blade.php; resources/css/app.css	Reuse semantic controls, navigation, and status components.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/edit.png	Verified

Nielsen's 10 Usability Heuristics

Table 8. Nielsen's 10 Usability Heuristics HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
H1	Visibility of system status	resources/views/layouts/app.blade.php; resources/views/portfolios/show.blade.php	Announce saved and published status and show the completion step.	docs/screenshots/mobile/step-3-done-share.png; PortfolioRoutesTest::test_template_selection_uses_and_preserves_the_private_link_for_the_creator	Verified
H2	Match between system and real world	resources/views/portfolios/form.blade.php; resources/views/help.blade.php	Use plain labels such as Skills, Projects, Save, and Next, with concrete examples.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/step-1-your-info.png	Verified
H3	User control and freedom	routes/web.php; resources/views/portfolios/deleted.blade.php; resources/js/app.js	Support back navigation, template changes, and deletion undo.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; docs/screenshots/mobile/delete-undo.png	Verified
H4	Consistency and standards	resources/views/layouts/app.blade.php; resources/css/app.css	Keep page structure, focus states, and control roles consistent.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/template-simple.png	Verified
H5	Error prevention	app/Http/Requests/Concerns/HasPortfolioRules.php; app/Http/Requests/SaveTemplateRequest.php	Validate required values, URL schemes, and supported templates before saving.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; PortfolioRoutesTest::test_template_choice_is_limited_to_the_three_supported_values	Verified
H6	Recognition rather than recall	resources/views/portfolios/select-template.blade.php; resources/views/portfolios/form.blade.php	Preview each template with the person's own information and show field examples.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/mobile/template-modern.png	Verified
H7	Flexibility and efficiency of use	resources/js/app.js; resources/views/layouts/app.blade.php	Provide keyboard shortcuts alongside standard pointer controls.	docs/screenshots/mobile/keyboard-shortcuts.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified
H8	Aesthetic and minimalist design	resources/views/portfolios/form.blade.php; resources/css/app.css	Show only the current creation task and group related information.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/template-simple.png	Verified
H9	Help users recognize, diagnose, recover from errors	resources/views/errors/status.blade.php; resources/js/app.js; app/Http/Requests/Concerns/HasPortfolioRules.php	Show friendly server errors and specific inline field guidance.	docs/screenshots/mobile/error-403.png; docs/screenshots/mobile/error-500.png; PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls	Verified
H10	Help and documentation	resources/views/help.blade.php; resources/views/layouts/app.blade.php	Link a help page and shortcut dialog from every screen.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/keyboard-shortcuts.png	Verified

Universal Design

Table 9. Universal Design HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
U1	Equitable use	resources/views/layouts/app.blade.php; resources/css/app.css	Provide keyboard and screen-reader access to core tasks.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/keyboard-shortcuts.png	Verified
U2	Flexibility in use	resources/js/app.js; resources/views/layouts/app.blade.php	Offer shortcuts and ordinary pointer controls.	docs/screenshots/mobile/keyboard-shortcuts.png; docs/screenshots/desktop/keyboard-shortcuts.png	Verified
U3	Simple and intuitive use	resources/views/portfolios/create.blade.php; resources/views/portfolios/select-template.blade.php; resources/views/portfolios/show.blade.php	Label the path from portfolio information through template choice to sharing.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/step-2-choose-template.png	Verified
U4	Perceptible information	resources/css/app.css; resources/views/portfolios/form.blade.php	Use contrast-tested text and pair status colors with readable labels.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/step-3-done-share.png	Verified
U5	Tolerance for error	routes/web.php; resources/views/portfolios/deleted.blade.php	Keep deleted content recoverable for 10 seconds, then permanently remove it.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; PortfolioModelTest::test_purge_command_removes_portfolios_after_the_undo_period; docs/screenshots/mobile/delete-undo.png	Verified
U6	Low physical effort	resources/css/app.css; resources/js/app.js	Support keyboard navigation and give controls generous spacing.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/keyboard-shortcuts.png	Verified
U7	Size and space for approach and use	resources/css/app.css	Give interactive controls at least 44 CSS pixels and adapt layouts for narrow screens.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/step-1-your-info.png	Verified

Color theory

Table 10. Color theory HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
C1	Deliberate color scheme per template	resources/css/app.css; docs/DECISIONS.md	Use monochromatic Simple neutrals, analogous blue Modern hues, and complementary plum/magenta with warm cream for Creative.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-modern.png; docs/screenshots/mobile/template-creative.png	Verified
C2	Color temperature matches the template's message	resources/css/app.css; docs/DECISIONS.md	Keep Simple restrained, Modern cool and calm, and Creative warm and expressive.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-modern.png; docs/screenshots/mobile/template-creative.png	Verified
C3	Sufficient value contrast, WCAG AA	resources/css/app.css; tests/Unit/ColorContrastTest.php	Check text pairs at 4.5:1 and control/focus boundaries at 3:1 in light and dark palettes.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/dark-mode-creative.png	Verified
C4	Never rely on color alone	resources/views/portfolios/form.blade.php; resources/views/portfolios/deleted.blade.php; resources/css/app.css	Pair completion, error, success, and undo states with visible text.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; docs/screenshots/mobile/error-429.png; docs/screenshots/mobile/step-3-done-share.png	Verified
C5	Define all colors as design tokens in one place	resources/css/app.css	Define app and template palette values in one root token block with dark overrides.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; resources/css/app.css	Verified

Visual design principles

Table 11. Visual design principles HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
V1	Unity	resources/css/app.css; resources/views/layouts/app.blade.php	Reuse typography, spacing, and component styles throughout the app.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/template-simple.png	Verified
V2	Gestalt principles	resources/css/app.css; resources/views/portfolios/form.blade.php	Group related fields, repeat section styling, and align content.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/template-creative.png	Verified
V3	Visual hierarchy	resources/css/app.css; resources/views/portfolios/stepper.blade.php	Use heading scale, step state, and spacing to guide the current task.	docs/screenshots/mobile/step-2-choose-template.png	Verified
V4	Balance	resources/css/app.css; resources/views/portfolios/templates/simple.blade.php	Balance primary content, supporting hints, and whitespace at each viewport.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-creative.png	Verified
V5	Contrast	resources/css/app.css; tests/Unit/ColorContrastTest.php	Separate text, controls, and focus states with measured contrast.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/dark-mode-creative.png	Verified
V6	Scale	resources/css/app.css; resources/views/portfolios/templates/creative.blade.php	Use a consistent readable type and component scale with responsive headings.	docs/screenshots/mobile/template-creative.png; docs/screenshots/desktop/template-creative.png	Verified
V7	Dominance	resources/css/app.css; resources/views/portfolios/create.blade.php	Make the current heading and primary action the clearest elements.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/mobile/step-1-your-info.png	Verified
V8	Similarity	resources/css/app.css; resources/views/layouts/app.blade.php	Style controls with the same role consistently throughout the application.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/edit.png	Verified

User needs

Table 12. User needs HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
R1	Documented personas and tasks	docs/USERS_AND_NEEDS.md	Describe student, freelancer, and job seeker contexts and tasks.	HciTrackerTest::test_documented_personas_and_core_tasks_cover_target_users; docs/USERS_AND_NEEDS.md	Verified
R2	Documented assumptions to validate	docs/USERS_AND_NEEDS.md	Separate assumptions from measured user research.	HciTrackerTest::test_user_research_assumptions_are_labeled_as_unvalidated; docs/USERS_AND_NEEDS.md	Verified
R3	Iteration log of changes from evaluation	docs/USERS_AND_NEEDS.md	Record evaluation evidence, findings, design changes, and follow-up work.	HciTrackerTest::test_iteration_log_records_evaluation_findings_and_design_changes; docs/USERS_AND_NEEDS.md; docs/COGNITIVE_WALKTHROUGH.md	Verified

7. Evaluation

7.1 Heuristic review

The single-reviewer expert inspection covers the 17 captured states against all ten Nielsen heuristics at mobile and desktop sizes. The screen-by-heuristic matrix, severity scale, findings, re-evaluation, and limitations are recorded in `docs/HEURISTIC_EVALUATION.md`. One minor finding remains: ten seconds is a short recovery period. A second cleanup finding was corrected by adding server-side scheduled purge and treating an already-purged record as completed. No severity 3 or 4 finding remains.

Table 13. Expert review findings and disposition

Finding	Severity	Disposition
The ten-second delete undo can be missed by a user who needs more time.	2	Retained to match the product requirement; ask classmates whether it is long enough.
Closing the delete screen could leave cleanup dependent on a later browser request.	2 before fix	Resolved with ten-second server-side purge scheduling and already-purged response handling; covered by purge command and schedule tests.

7.2 Cognitive walkthrough

This walkthrough applies the student, freelancer, and job-seeker personas to the implemented task path. It is an expert review based on the screens and route behavior, not observed participant behavior. The step-by-step record is in `docs/COGNITIVE_WALKTHROUGH.md`.

Table 14. Expert cognitive walkthrough

Persona	Task path reviewed	Expected user questions	Inspection result and limitation
Student	Add name, headline, introduction, skills, and optional education or course projects; select a template; copy the public URL.	“What information is required?”, “Can I leave work experience empty?”, “How do I share?”	Field guidance, optional section labels, a required skill message, template previews, and a completion screen answer the main questions. A classmate should validate terminology and whether the required fields feel reasonable.
Freelancer	Add a specialty and projects, preview styles, publish, save the private management link, and later update a project.	“Will my work look professional?”, “Which link is public?”, “Can I edit after sharing?”	The three previews use the same profile data; public and private links have separate labels; the private link opens edit and template controls. The owner must retain the signed link because there is no account recovery.
Job seeker	Add experience and skills, use keyboard navigation, publish a focused page, and test deletion recovery.	“Where am I in the process?”, “Can I correct a mistake?”, “Can I undo a delete?”	The stepper, field validation, keyboard shortcuts, edit page, and ten-second undo provide visible controls. Ten seconds may be too brief for some users and is an explicit study question.

7.3 Usability test plan

The kit in `docs/usability-testing/` contains a five-task moderator script, an observation sheet, a ten-item SUS questionnaire, and a results table. Participant count, task results, quotes, and SUS values remain marked “to be completed with classmates.” No user-study score is reported in this paper.

7.4 Iteration record

The R3 log in `docs/USERS_AND_NEEDS.md` records the draft restoration and repeat-row fixes observed during browser testing, and the server-side purge issue found during code inspection. The log distinguishes automation evidence from future classmate evaluation.

8. Deployment and infrastructure

8.1 Release configuration

`Dockerfile` uses Composer, Node, and PHP 8.3 Apache stages. The final image installs PostgreSQL support, serves `public/`, reads the platform `PORT`, runs additive migrations and the idempotent demo seed, caches Laravel configuration/routes/views, and supervises Apache with `schedule:work`. `render.yaml` defines a Render Docker web service, `/up` health check, `checksPass` deployment trigger, and environment variable names with secret values supplied out of band. `.github/workflows/smoke.yml` checks a configured live URL after CI.

Figure 37. CI and deployment flow



Table 15. Deployment environment variables

Variable	Value source
APP_ENV, APP_DEBUG, APP_TIMEZONE	Render Blueprint settings; production, false, and Asia/Taipei.
APP_KEY	Generate and store in Render secret environment.
APP_URL	HTTPS origin of the deployed Render service.
DB_CONNECTION, DB_PORT, DB_DATABASE, DB_SSLMODE	PostgreSQL, 5432, postgres (or configured DB name), and require.
DB_HOST	Supabase shared session-pooler host copied from its Connect screen.
DB_USERNAME	Supabase session-pooler username, typically postgres.[PROJECT_REF].
DB_PASSWORD	Supabase database password stored as a Render secret.
SESSION_DRIVER, CACHE_STORE, QUEUE_CONNECTION	File, file, and sync for the single-service configuration.
PORTGEN_LIVE_URL	GitHub repository variable for the smoke workflow, set only after a real deployment.

Supabase session pooler connectivity is configured for port 5432 and SSL mode `require`. Production migrations must be additive; `migrate:fresh` and `db:wipe` are reserved for the isolated CI database. The local test guard refuses a Supabase host. The project currently has no production database configuration in this checkout.

8.2 Deployment status and blockers

The GitHub and Supabase browser sessions are signed into the account whose name begins with `crazyfrog`, but this checkout has no Git remote, GitHub CLI, Docker, Render API key, Supabase database host, username, or password. The local `portgen-secrets.env` file is absent. The visible Supabase organization already contains another project; it was not modified. No PortGen Supabase project or Render service was created.

Laravel's published release table shows that Laravel 11 security fixes ended on 2026-03-12. The framework remains Laravel 11 because the approved brief fixes that stack. Composer reports four advisories for the current framework dependency. Therefore no public deployment or production database migration was performed. Resolve the framework

support issue, then provide the service configuration and run the PostgreSQL CI, Docker build, Render deploy, and live smoke workflow.

9. Testing and technical achievements

9.1 Local command results

Table 16. Verification executed in this workspace

Command	Result
php artisan test --compact	Exit code 0; 573 assertions; 37 deprecation-marked tests and 1 passing test in the compact PHP 8.5 output.
./vendor/bin/pint --test	Passed.
npm run build	Passed with Vite 8.3.4.
npm audit --omit=optional	Zero vulnerabilities.
npm run test:e2e	1 Playwright test passed; captured 17 states at mobile and desktop sizes.
php artisan schedule:list	Shows portfolios:purge-expired every ten seconds.
HCI_RELEASE=1 php artisan test --compact	Exit code 0; 657 assertions; the tracker release requirement passes. PHP 8.5 reports 37 deprecation-marked tests and 1 passing test in the compact summary.
composer audit --locked	Exit code 1; four Laravel framework advisories found, including a medium-severity signed-URL path-confusion advisory.

The workstation uses PHP 8.5, not the required PHP 8.3 runtime, so PHPUnit reports deprecations. CI is configured to use PHP 8.3 and PostgreSQL 16. PostgreSQL migration rollback/reapply, CI, and a Docker image build were not run here.

9.2 Database, browser, and security checks

Feature tests cover portfolio validation, signed access, template constraints, escaping, rate limits, demo seeding, deletion/undo, and purge behavior. Model tests cover UUID and slug behavior, mass-assignment protection, uniqueness, JSON arrays, template constraints, seeder idempotence, and scheduled cleanup. Browser checks use a dedicated SQLite test database and test keyboard flow, draft state, responsive widths, dark mode, and error screens. CI declares an isolated PostgreSQL 16 database and an explicit test-only guard; these PostgreSQL jobs remain configured but unexecuted on the workstation.

The implementation includes CSRF, rate limits on state-changing routes, signed owner URLs, strict URL schemes, escaped user content, friendly error views, security headers, an explicit mass-assignment allowlist, and a repository secret scanner. The signed owner URL is a bearer credential and should be kept private. No live proxy response, including a 502, was available to inspect because no service is deployed; the smoke workflow is configured to check the service endpoints after a future deployment.

10. User guide

10.1 Create and publish

1. Open `/portfolios/create` and enter your name, headline, and introduction.
2. Expand the optional Education, Experience, Projects, and Links sections as needed. Add at least one skill.
3. Use Next to compare previews populated with your information.
4. Choose Simple, Modern, or Creative, then finish the sharing step.
5. Copy the public URL to share your portfolio. Save the separate private management link in a secure place.

10.2 Manage and edit

Open the private signed link to view the portfolio controls. Choose Edit portfolio to update details or Change template to preview another style. Anyone who has the private link can manage the portfolio, so do not share it publicly.

10.3 Delete and undo

Choose Delete portfolio and review the named confirmation. The deletion screen provides a ten-second Undo control. After the window ends, the server scheduler permanently removes the record. The local copy in the browser is not a backup.

10.4 Keyboard shortcuts

Use Ctrl/Cmd+S to save a draft or submit the current step, Alt+Right/Left to move through steps, and `?` to open the shortcut dialog. The shared layout also links the Help and FAQ page.

11. Maintenance and future work

11.1 Run locally

1. Install PHP 8.3, Composer, Node.js 24, and npm.
2. Run `composer install` and `npm ci`.
3. Copy `.env.example` to `.env`, generate an application key with `php artisan key:generate`, and configure an isolated local database.
4. Run `php artisan migrate --seed` and `npm run build`.
5. Start the app with `php artisan serve` and the front-end watcher with `npm run dev`.
6. Run `php artisan test`, `./vendor/bin/pint --test`, `npm audit --omit=optional`, and `npm run test:e2e` for the local checks.

Never point test commands or `migrate:fresh` at Supabase. For a supported public launch, upgrade the framework dependency to a currently supported Laravel release and re-run the full CI gate.

11.2 Future work

- Upgrade from Laravel 11 to a supported framework release while preserving behavior and retesting dependencies.
- Connect the repository to GitHub and run the CI workflow against PostgreSQL 16.
- Build the Docker image, configure a new PortGen database and Render service, then pass live smoke checks.
- Conduct the five-task classmate study and add real observation and SUS results.
- Consider profile photos and project images only after launch readiness; validate type, size, alt text, preview, storage, and deletion flows.
- Evaluate skill-logo suggestions with an explicit text fallback and no dependency on color or imagery.

12. Appendices

Appendix A. Decisions

Table 17. Dated implementation decisions

Date	Decision	Reason
2026-10-19	Keep Laravel 11 as the implementation target because the approved product brief fixes that stack; use the latest 11.x line available from Composer.	Changing the framework major would conflict with the explicit stack requirement. Composer's current security policy blocked normal dependency resolution, so the scaffold was installed with --no-security-blocking; Composer reported four advisories affecting laravel/framework. Laravel's published support table ends Laravel 11 security fixes on 2026-03-12. Treat production deployment as blocked until the framework is upgraded or the risk is explicitly resolved; do not represent this stack as production-safe.
2026-10-19	Set Composer's platform PHP version to 8.3.	The required CI and production runtime are PHP 8.3, while the workstation currently runs PHP 8.5. This keeps dependency resolution compatible with the requested runtime.
2026-10-19	Use Tailwind CSS 4 through its official Vite plugin and use Playwright with axe-core for browser checks.	These match the required build and browser-testing stack; the packages are dev dependencies and are needed for the requested UI and accessibility gates.
2026-10-19	Upgrade Vite and laravel-vite-plugin to versions compatible with Node 24 and Vite 8 after npm audit found vulnerable esbuild/Vite dependencies.	The existing Laravel 11 skeleton pinned Vite 5, which produced moderate and high npm audit findings. The current Laravel Vite plugin supports Vite 8; the updated dependency tree reports zero npm vulnerabilities.
2026-10-19	Use neutral monochrome tokens for Simple, cool neighboring blue hues for Modern, and warm cream with plum/magenta accents for Creative.	The palettes distinguish quiet, focused, and expressive presentation styles while keeping text and controls readable in both color schemes; ColorContrastTest measures the defined foreground/background pairs.
2026-10-19	Use a deterministic fictional public demo slug (portgen001) and an idempotent seeder for the live smoke check.	The deployment smoke workflow needs a stable public URL to verify that migrations, seeding, and public rendering work together without using real user data.
2026-10-19	Keep local error-preview routes available only in local and testing environments.	Playwright needs to inspect every friendly server-error screen, while production should expose only application routes.
2026-10-19	Run local Playwright with a dedicated temporary SQLite file and reserve PostgreSQL browser tests for the isolated CI service.	This machine has no Docker, PostgreSQL server, or GitHub runner access. The browser setup overrides DB connection settings and refuses a PostgreSQL target unless it is explicitly marked as the local portgen_test service.
2026-10-19	Do not deploy the Laravel 11 dependency set while its security support window is closed.	The required stack fixes Laravel 11, but its published security-fix date has passed and Composer reports four advisories. The deployment files are prepared, but a live service is not claimed safe.
2026-10-19	Use SQLite in memory only for the initial scaffold checks; use the dedicated local Postgres 16 service for application database tests and CI.	This avoids any accidental connection to production and allows the scaffold gate to run before a Docker service is available.

Date	Decision	Reason
2026-10-19	Run the required Postgres 16 migration, rollback, and feature gates in the CI service; run only SQLite-backed migration/model tests on this workstation.	Docker, psql, and a local PostgreSQL service are absent, and the current Windows account cannot install a PostgreSQL service. The CI job declares its own ephemeral Postgres 16 service. Local SQLite results do not count as the Postgres gate.
2026-10-19	Run expired portfolio cleanup every ten seconds and supervise Laravel's scheduler with Apache in the container.	The required ten-second undo period must also finish when an owner closes the browser. Laravel supports sub-minute schedules through schedule:work; a test covers the purge command and schedule entry.
2026-10-19	Keep the usability study kit open for classmates and report no participant scores until sessions have been run.	The implementation has expert inspection and automated checks, but no real participant data. Blank observation and results fields prevent fabricated findings.
2026-10-19	Store Mermaid source and a neutral theme configuration next to rendered diagrams and use the diagrams in the system paper.	Source files keep architecture, data, and release flow editable; rendered PNGs allow diagrams to be embedded in the DOCX and PDF. Mermaid CLI is used only as an authoring tool and is not added to application dependencies.

Appendix B. Glossary

Bearer link: A private URL whose possession grants access to a resource.

JSONB: PostgreSQL's binary JSON data type for structured values.

Signed URL: A URL with a cryptographic signature that detects tampering.

Soft delete: Marking a row as deleted while retaining it for a recovery period.

SUS: System Usability Scale, a standardized ten-item questionnaire.

WCAG: Web Content Accessibility Guidelines published by W3C.

Appendix C. References

1. Shneiderman, B., Plaisant, C., Cohen, M., Jacobs, S., and Elmqvist, N. *Designing the User Interface: Strategies for Effective Human-Computer Interaction*, 6th ed., Pearson, 2016. Summary of the Eight Golden Rules: University of Maryland HCIL.
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3. Nielsen, J. and Molich, R. “Heuristic Evaluation of User Interfaces.” *Proceedings of CHI 1990*, pp. 249-256. doi:10.1145/97243.97281.
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10. Render. “Blueprint Specification Reference.” Official documentation.
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Appendix D. HCI tracker copy

This copy matches the 48 implementation rows in section 6.4 and `docs/HCI_TRACKER.md`.

Shneiderman's 8 Golden Rules

Table 18. Shneiderman's 8 Golden Rules HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
S1	Consistency	resources/views/layouts/app.blade.php; resources/css/app.css	Shared layout, navigation, spacing, and repeated button styles.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/desktop/step-1-your-info.png	Verified
S2	Shortcuts for frequent users	resources/js/app.js; resources/views/layouts/app.blade.php	Discoverable save and step navigation shortcuts.	docs/screenshots/mobile/keyboard-shortcuts.png; docs/screenshots/desktop/keyboard-shortcuts.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified
S3	Informative feedback	resources/views/layouts/app.blade.php; resources/views/portfolios/show.blade.php	Announce saved, published, copied-link, and error status in polite live regions.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/desktop/step-3-done-share.png; PortfolioRoutesTest::test_template_selection_uses_and_preserves_the_private_link_for_the_creator	Verified
S4	Dialogs that yield closure	resources/views/portfolios/manage.blade.php; resources/views/portfolios/show.blade.php	Name the portfolio in the delete dialog and show the private owner link when creation finishes.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/desktop/step-3-done-share.png	Verified
S5	Error prevention and simple error handling	app/Http/Requests/Concerns/HasPortfolioRules.php; resources/views/portfolios/form.blade.php; resources/js/app.js	Validate structured fields and safe URLs, then show specific inline recovery messages.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; docs/screenshots/mobile/step-1-your-info.png	Verified
S6	Easy reversal of actions	routes/web.php; resources/views/portfolios/deleted.blade.php; app/Http/Controllers/PortfolioController.php	Offer a signed 10-second undo before permanent removal.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; docs/screenshots/mobile/delete-undo.png; docs/screenshots/desktop/delete-undo.png	Verified
S7	Internal locus of control	routes/web.php; app/Http/Controllers/PortfolioController.php	Require a private signed URL for management and keep transitions user initiated.	PortfolioRoutesTest::test_signed_manage_link_opens_and_unsigned_link_is_forbidden; docs/screenshots/mobile/manage.png	Verified
S8	Reduce short-term memory load	resources/views/portfolios/form.blade.php; resources/js/app.js	Group form fields, show completion labels, and offer local draft restoration.	docs/screenshots/mobile/step-1-your-info.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified

Norman's 7 Principles

Table 19. Norman's 7 Principles HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
N1	Knowledge in the world and in the head	resources/views/portfolios/form.blade.php; resources/views/help.blade.php	Use examples and plain-language help beside unfamiliar fields.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/step-1-your-info.png	Verified
N2	Simplify task structure	resources/views/portfolios/stepper.blade.php; resources/views/portfolios/create.blade.php; resources/views/portfolios/select-template.blade.php	Divide creation into a visible three-step flow.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/desktop/step-2-choose-template.png	Verified
N3	Make things visible	resources/views/portfolios/stepper.blade.php; resources/js/app.js; resources/views/layouts/app.blade.php	Show current step, section completion, save status, and submit loading state.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/mobile/step-3-done-share.png	Verified
N4	Get the mappings right	resources/views/portfolios/form.blade.php; resources/views/portfolios/templates/sections.blade.php	Use familiar field names and group entered information under matching headings.	docs/screenshots/mobile/step-1-your-info.png	Verified
N5	Use constraints	app/Http/Requests/SaveTemplateRequest.php; app/Http/Requests/Concerns/HasPortfolioRules.php	Restrict template choice and URL schemes before storage.	PortfolioRoutesTest::test_template_choice_is_limited_to_the_three_supported_values; docs/screenshots/mobile/step-2-choose-template.png	Verified
N6	Design for error	resources/js/app.js; resources/views/portfolios/deleted.blade.php	Autosave drafts, explain validation failures, and provide delete recovery.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window	Verified
N7	When all else fails, standardize	resources/views/layouts/app.blade.php; resources/css/app.css	Reuse semantic controls, navigation, and status components.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/edit.png	Verified

Nielsen's 10 Usability Heuristics

Table 20. Nielsen's 10 Usability Heuristics HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
H1	Visibility of system status	resources/views/layouts/app.blade.php; resources/views/portfolios/show.blade.php	Announce saved and published status and show the completion step.	docs/screenshots/mobile/step-3-done-share.png; PortfolioRoutesTest::test_template_selection_uses_and_preserves_the_private_link_for_the_creator	Verified
H2	Match between system and real world	resources/views/portfolios/form.blade.php; resources/views/help.blade.php	Use plain labels such as Skills, Projects, Save, and Next, with concrete examples.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/step-1-your-info.png	Verified
H3	User control and freedom	routes/web.php; resources/views/portfolios/deleted.blade.php; resources/js/app.js	Support back navigation, template changes, and deletion undo.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; docs/screenshots/mobile/delete-undo.png	Verified
H4	Consistency and standards	resources/views/layouts/app.blade.php; resources/css/app.css	Keep page structure, focus states, and control roles consistent.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/template-simple.png	Verified
H5	Error prevention	app/Http/Requests/Concerns/HasPortfolioRules.php; app/Http/Requests/SaveTemplateRequest.php	Validate required values, URL schemes, and supported templates before saving.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; PortfolioRoutesTest::test_template_choice_is_limited_to_the_three_supported_values	Verified
H6	Recognition rather than recall	resources/views/portfolios/select-template.blade.php; resources/views/portfolios/form.blade.php	Preview each template with the person's own information and show field examples.	docs/screenshots/mobile/step-2-choose-template.png; docs/screenshots/mobile/template-modern.png	Verified
H7	Flexibility and efficiency of use	resources/js/app.js; resources/views/layouts/app.blade.php	Provide keyboard shortcuts alongside standard pointer controls.	docs/screenshots/mobile/keyboard-shortcuts.png; tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible	Verified
H8	Aesthetic and minimalist design	resources/views/portfolios/form.blade.php; resources/css/app.css	Show only the current creation task and group related information.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/template-simple.png	Verified
H9	Help users recognize, diagnose, recover from errors	resources/views/errors/status.blade.php; resources/js/app.js; app/Http/Requests/Concerns/HasPortfolioRules.php	Show friendly server errors and specific inline field guidance.	docs/screenshots/mobile/error-403.png; docs/screenshots/mobile/error-500.png; PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls	Verified
H10	Help and documentation	resources/views/help.blade.php; resources/views/layouts/app.blade.php	Link a help page and shortcut dialog from every screen.	docs/screenshots/mobile/help-faq.png; docs/screenshots/mobile/keyboard-shortcuts.png	Verified

Universal Design

Table 21. Universal Design HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
U1	Equitable use	resources/views/layouts/app.blade.php; resources/css/app.css	Provide keyboard and screen-reader access to core tasks.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/keyboard-shortcuts.png	Verified
U2	Flexibility in use	resources/js/app.js; resources/views/layouts/app.blade.php	Offer shortcuts and ordinary pointer controls.	docs/screenshots/mobile/keyboard-shortcuts.png; docs/screenshots/desktop/keyboard-shortcuts.png	Verified
U3	Simple and intuitive use	resources/views/portfolios/create.blade.php; resources/views/portfolios/select-template.blade.php; resources/views/portfolios/show.blade.php	Label the path from portfolio information through template choice to sharing.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/step-2-choose-template.png	Verified
U4	Perceptible information	resources/css/app.css; resources/views/portfolios/form.blade.php	Use contrast-tested text and pair status colors with readable labels.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/step-3-done-share.png	Verified
U5	Tolerance for error	routes/web.php; resources/views/portfolios/deleted.blade.php	Keep deleted content recoverable for 10 seconds, then permanently remove it.	PortfolioRoutesTest::test_delete_can_be_undone_during_the_ten_second_window; PortfolioModelTest::test_purge_command_removes_portfolios_after_the_undo_period; docs/screenshots/mobile/delete-undo.png	Verified
U6	Low physical effort	resources/css/app.css; resources/js/app.js	Support keyboard navigation and give controls generous spacing.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/keyboard-shortcuts.png	Verified
U7	Size and space for approach and use	resources/css/app.css	Give interactive controls at least 44 CSS pixels and adapt layouts for narrow screens.	tests/e2e/portfolio-accessibility.spec.js::keyboard-first create, publish, manage, template, and delete-undo flows stay accessible; docs/screenshots/mobile/step-1-your-info.png	Verified

Color theory

Table 22. Color theory HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
C1	Deliberate color scheme per template	resources/css/app.css; docs/DECISIONS.md	Use monochromatic Simple neutrals, analogous blue Modern hues, and complementary plum/magenta with warm cream for Creative.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-modern.png; docs/screenshots/mobile/template-creative.png	Verified
C2	Color temperature matches the template's message	resources/css/app.css; docs/DECISIONS.md	Keep Simple restrained, Modern cool and calm, and Creative warm and expressive.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-modern.png; docs/screenshots/mobile/template-creative.png	Verified
C3	Sufficient value contrast, WCAG AA	resources/css/app.css; tests/Unit/ColorContrastTest.php	Check text pairs at 4.5:1 and control/focus boundaries at 3:1 in light and dark palettes.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/dark-mode-creative.png	Verified
C4	Never rely on color alone	resources/views/portfolios/form.blade.php; resources/views/portfolios/deleted.blade.php; resources/css/app.css	Pair completion, error, success, and undo states with visible text.	PortfolioRoutesTest::test_store_requires_a_skill_and_rejects_unsafe_urls; docs/screenshots/mobile/error-429.png; docs/screenshots/mobile/step-3-done-share.png	Verified
C5	Define all colors as design tokens in one place	resources/css/app.css	Define app and template palette values in one root token block with dark overrides.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; resources/css/app.css	Verified

Visual design principles

Table 23. Visual design principles HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
V1	Unity	resources/css/app.css; resources/views/layouts/app.blade.php	Reuse typography, spacing, and component styles throughout the app.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/template-simple.png	Verified
V2	Gestalt principles	resources/css/app.css; resources/views/portfolios/form.blade.php	Group related fields, repeat section styling, and align content.	docs/screenshots/mobile/step-1-your-info.png; docs/screenshots/mobile/template-creative.png	Verified
V3	Visual hierarchy	resources/css/app.css; resources/views/portfolios/stepper.blade.php	Use heading scale, step state, and spacing to guide the current task.	docs/screenshots/mobile/step-2-choose-template.png	Verified
V4	Balance	resources/css/app.css; resources/views/portfolios/templates/simple.blade.php	Balance primary content, supporting hints, and whitespace at each viewport.	docs/screenshots/mobile/template-simple.png; docs/screenshots/mobile/template-creative.png	Verified
V5	Contrast	resources/css/app.css; tests/Unit/ColorContrastTest.php	Separate text, controls, and focus states with measured contrast.	ColorContrastTest::test_light_and_dark_design_tokens_meet_wcag_contrast_thresholds; docs/screenshots/mobile/dark-mode-creative.png	Verified
V6	Scale	resources/css/app.css; resources/views/portfolios/templates/creative.blade.php	Use a consistent readable type and component scale with responsive headings.	docs/screenshots/mobile/template-creative.png; docs/screenshots/desktop/template-creative.png	Verified
V7	Dominance	resources/css/app.css; resources/views/portfolios/create.blade.php	Make the current heading and primary action the clearest elements.	docs/screenshots/mobile/step-3-done-share.png; docs/screenshots/mobile/step-1-your-info.png	Verified
V8	Similarity	resources/css/app.css; resources/views/layouts/app.blade.php	Style controls with the same role consistently throughout the application.	docs/screenshots/mobile/manage.png; docs/screenshots/mobile/edit.png	Verified

User needs

Table 24. User needs HCI tracker

ID	Principle	Where applied	How applied	Evidence	Status
R1	Documented personas and tasks	docs/USERS_AND_NEEDS.md	Describe student, freelancer, and job seeker contexts and tasks.	HciTrackerTest::test_documented_personas_and_core_tasks_cover_target_users; docs/USERS_AND_NEEDS.md	Verified
R2	Documented assumptions to validate	docs/USERS_AND_NEEDS.md	Separate assumptions from measured user research.	HciTrackerTest::test_user_research_assumptions_are_labeled_as_unvalidated; docs/USERS_AND_NEEDS.md	Verified
R3	Iteration log of changes from evaluation	docs/USERS_AND_NEEDS.md	Record evaluation evidence, findings, design changes, and follow-up work.	HciTrackerTest::test_iteration_log_records_evaluation_findings_and_design_changes; docs/USERS_AND_NEEDS.md; docs/COGNITIVE_WALKTHROUGH.md	Verified